3. $p(x) = \begin{cases} \frac{e^{-\frac{x}{\theta}}}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases}$, $\theta > 0$ - ВМ.

$n = 3$

Оценки:

$\tilde{\theta}_1 = \bar{x}$ $\quad \tilde{\theta}_2 = x_{(2)}$

а) $\forall\theta \; M\tilde{\theta}_1 = Mx = \theta$ - неим.

$$\forall\theta \; M\tilde{\theta}_2 = \int_0^\infty x \frac{n(n-1)}{\theta}\left(e^{-(n-1)\frac{x}{\theta}} - e^{-n\frac{x}{\theta}}\right)dx =$$

$$= \int_0^\infty x \frac{3\cdot 2}{\theta} e^{-2\frac{x}{\theta}} dx - \int_0^\infty x \frac{3\cdot 2}{\theta} e^{-3\frac{x}{\theta}} dx =$$

$$= \frac{3\theta}{2}\underbrace{\int_0^\infty \frac{2x}{\theta} e^{-\frac{2x}{\theta}} d\frac{2x}{\theta}}_{1} - \frac{2}{3}\theta\underbrace{\int_0^\infty \frac{3x}{\theta} e^{-\frac{3x}{\theta}} d\frac{3x}{\theta}}_{1} = \frac{9\theta - 4\theta}{6} = \frac{5}{6}\theta \neq \theta \text{ - смещ}$$

$\Rightarrow \tilde{\theta}_2^* = \frac{6}{5}\theta$ - неимещ., исправл.

б) • Регулярность модели

1) $\frac{\partial}{\partial\theta} F(x,\theta) = \frac{\partial}{\partial\theta}\left(1 - e^{-\frac{x}{\theta}}\right) = -e^{-\frac{x}{\theta}} \cdot \left(\frac{x}{\theta^2}\right)$, $x \geq 0$

$\frac{\partial}{\partial\theta} F(x,\theta) = 0$, $x \leq 0$

$\Big\}$ $\theta > 0$ - Θ

$\Rightarrow \frac{\partial}{\partial\theta} F(x,\theta)$ - непрер. по θ на Θ. ✓

г) $$-\int_0^\infty d\frac{x}{\theta^2} e^{-\frac{x}{\theta}} = -\int_0^\infty \frac{1}{\theta^2} e^{-\frac{x}{\theta}} + \frac{x}{\theta^2} e^{-\frac{x}{\theta}} \cdot \left(-\frac{1}{\theta}\right) dx =$$

$$= -\frac{1}{\theta^2}\int_0^\infty \left(e^{-\frac{x}{\theta}} - \frac{x}{\theta} e^{-\frac{x}{\theta}}\right) dx = -\frac{1}{\theta^2}\left[\theta - \theta\right] = 0 \; ✓$$

3) $$I(\theta) = \int_0^\infty \left(\frac{\partial}{\partial\theta}\left(-\ln\theta - \frac{x}{\theta}\right)\right)^2 \frac{e^{-\frac{x}{\theta}}}{\theta} dx = \int_0^\infty \left(-\frac{1}{\theta} + \frac{x}{\theta^2}\right)^2 \frac{e^{-\frac{x}{\theta}}}{\theta} dx =$$

$$= \int_0^\infty \frac{1}{\theta^3}\left(1 - 2\frac{x}{\theta} + \frac{x^2}{\theta^2}\right) e^{-\frac{x}{\theta}} dx = \frac{1}{\theta^2}\underbrace{\int_0^\infty e^{-\frac{x}{\theta}} d\frac{x}{\theta}}_{1} - \frac{2}{\theta^2}\underbrace{\int_0^\infty \frac{x}{\theta} e^{-\frac{x}{\theta}} d\frac{x}{\theta}}_{1} +$$

$$+ \frac{1}{\theta^2}\underbrace{\int_0^\infty \frac{x^2}{\theta^2} e^{-\frac{x}{\theta}} d\frac{x}{\theta}}_{2} = \frac{2}{\theta^2} > 0$$ и непр. на Θ ✓

$\Rightarrow$ модель регулярна.

- Регулярность оценок.

Дост. усл. регуляр.

$\tilde{\theta}_1$: модель рег. ✓

оценка несмещ. ✓

$D\tilde{\theta}_1 = \frac{1}{n} Dx = \frac{\theta^2}{3}$ - огран. на $\forall$ знач. из Θ ✓

$\Rightarrow \tilde{\theta}_1$ - рег. ✓

$\tilde{\theta}_2^*$: модель рег. ✓

несм. ✓

$D\tilde{\theta}_2^* \approx 5{,}6\,\theta^2$ - огран. на $\forall$ знач. из Θ ✓

$\Rightarrow \tilde{\theta}_2^*$ - рег. ✓

т.к. $D\tilde{\theta}_1 < D\tilde{\theta}_2^*$, $\tilde{\theta}_1$ ближе в смысле $|D_{эфф} - D\tilde{\theta}_1|$ к эфф. оценке, чем $\tilde{\theta}_2$.

c) Нерав-во Кр.-Р.

модель рег. ✓

~~статистики~~ оценки рег. ✓ об эффект. ничего сказать нельзя.

$$\frac{(g'(\theta))^2}{nI(\theta)} = \frac{\theta^2}{n2} = \frac{\theta^2}{6} \Rightarrow$$ ~~ни $\tilde{\theta}_1$, ни $\tilde{\theta}_2^*$ не явл. эффект.~~

4. $\xi \sim \rho(x) = a\{(-1,1)\} + b\{2,0\}$.

$$\int_{-\infty}^{\infty} dF(x) = \int_{-1}^{1} a\,dx + 2b = 1$$

$$\frac{1}{2} a(1+1) + 2b = 1 \Rightarrow$$

$$\Rightarrow a = \frac{1}{2} - b.$$

$a = \theta$,

$b = \frac{1}{2} - \theta$, $\theta \in (0, \frac{1}{2})$ - Θ.

n - объём выборки.

а) 1. Оценка парам. модели методом ~~макс. правд.~~ моментов

$\xi \sim \rho(x) = \theta\{(-1,1)\} + (1-\theta)\{0,2\}$; $\theta \in \Theta = (0, \frac{1}{2})$ ВМ.

$\xrightarrow{} x_n$ $M\xi = \int_{-\infty}^{\infty} x\,dF(x) = \int_{-1}^{1} \theta x\,dx + 0\cdot(1-\theta) + 2\cdot(1-\theta) =$

$= \frac{1}{2}\theta(1-1) + 2(1-\theta) = 2 - 2\theta$. $\alpha_1(\theta) = \tilde{\alpha}_1$.

$$\boxed{\tilde{\theta}_1 = 1 - \frac{\overline{x}}{2}}$$

2. Оценка парам. методом макс. правд.

$\xi \sim \rho(x) = \theta\{(-1,1)\} + (1-\theta)\{0,2\}$; $\theta \in \Theta = (0, \frac{1}{2})$ ВМ.

$L(\theta) = \prod_{i=1}^{n} \rho(x_i, \theta) = \theta^m (1-\theta)^{n-m}$, m - число элем. из $(-1,1)\setminus\{0\}$

$\sup_{\theta \in \Theta} L(\theta) \to \tilde{\theta}_2 = \frac{m}{n}$, т.к. $L' = \theta^{m-1}(1-\theta)^{n-m-1}(m(1-\theta) - (n-m)\theta)$.

Проверка на max:

$L'' = [\theta^{m-1}(1-\theta)^{n-m-1}(m - n\theta)]' = (m-1)\theta^{m-2}(1-\theta)^{n-m-1}(m-n\theta) -$

$-(n-m-1)(1-\theta)^{n-m-2}\theta^{m-1}(m-n\theta) - n(1-\theta)^{n-m-1}\theta^{m-1} =$

$= \theta^{m-2}(1-\theta)^{n-m-2}\big((m-1)(1-\theta)(m-n\theta)-(n-m-1)\theta(m-n\theta)-$

$-n(1-\theta)\theta\big) = \theta^{m-2}(1-\theta)^{n-m-2}\big((m-m\theta-1+\theta)(m-n\theta)-(nm\theta-n^2\theta^2-m^2\theta+n\theta^2)-$

$-n\theta+n\theta^2\big) = \theta^{m-2}(1-\theta)^{n-m-2}\big(m^2-\cancel{m^2\theta}-m+m\theta-nm\theta+nm\theta^2+\cancel{n\theta}\ \cancel{n\theta^2}$

$-nm\theta+n^2\theta^2+\cancel{m^2\theta}-\cancel{n\theta^2}-\cancel{n\theta}+n\theta^2\big)=$

$(m^2-m+m\theta-2nm\theta+nm\theta^2+n^2\theta^2-n\theta^2)=$

$= m^2-m+\frac{m^2}{n}-2m^2+\frac{m^2}{n}+m^2-\frac{m^2}{n} = \frac{m^2-mn}{n} = \frac{m(m-n)}{n}<0$ ✓.

б) • Несм. $\tilde\theta_1$.

~~$M\xi = \int_{-\infty}^{\infty} x\,dF(x) =$~~

$\forall\theta: M\tilde\theta_1 = \cancel{\frac{3}{2}(2-M\xi)}\ 1-\frac{M\xi}{2} = \theta$. – несм. ✓

• Несм. $\tilde\theta_2$

$\forall\theta: M\tilde\theta_2 = \cancel{\frac{1}{n}\sum}\ M\xi = \theta$. – несм. ✓

~~c) Исслед~~

• Сост. $\tilde\theta_1$

$D\tilde\theta_1 = \frac{1}{4n^2}D\sum_{i=1}^{n}x_i = \frac{1}{4n}D\xi \xrightarrow[n\to\infty]{} 0$ – сост. по дост. усл. ✓

$M\xi^2 = \int_{-1}^{1}\theta x^2\,dx + 4(1-\theta) = \frac{2}{3}\theta + 4 - 4\theta = 4-\frac{10}{6}\theta$

$D\xi = \cancel{4}-\frac{5}{6}\theta-\cancel{4}+8\theta-4\theta^2 = \frac{43}{6}\theta-4\theta^2$ – конечн.

• Сост. $\tilde\theta_2$

$D\tilde\theta_2 = \frac{\theta(1-\theta)}{n} \xrightarrow[n\to\infty]{} 0$ – сост. по дост. усл. ✓

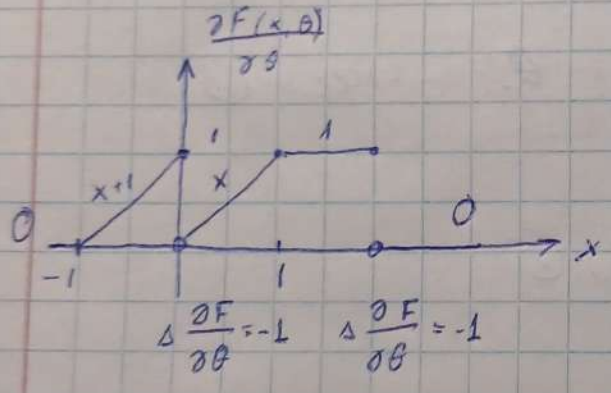
c) Исследование на эрф с нам-ю усл. Кр.-Р.

• $F(x,\theta) = \int_{-\infty}^{x} dF(x) = \cancel{\begin{cases} 0, & x\in(-\infty,-1) \\ \theta(x+1), & x\in[-1,0] \\ 0, \end{cases}}$

$$= \begin{cases} 0, & x\in(-\infty,-1] \\ \theta(x+1), & x\in[-1,0] \\ \frac{1}{2}+\theta(x), & x\in(0,1] \\ \frac{1}{2}-\theta, & x\in(1,2] \\ 1, & x\in(2,+\infty). \end{cases}$$

$F(x,\theta)$ – непрер. дифф. по θ. ✓

• $\int_{-\infty}^{\infty} d\frac{\partial F(x,\theta)}{\partial\theta} = \int_{-1}^{0} 1\,dx + \int_{0}^{1} 1\,dx + (-1)+(-1) = 0$ ✓



• $I(\theta) = \int_{-1}^{1}\left(\frac{\partial \ln\theta}{\partial\theta}\right)^2\theta\,dx + 2\left(\frac{\partial\ln(1-\theta)}{\partial\theta}\right)^2(1-\theta) =$

$= \int_{-1}^{1}\frac{1}{\theta^2}\theta\,dx + 2\frac{1}{(1-\theta)^2}(1-\theta) = \frac{2}{\theta}+\frac{2}{1-\theta}$ непрер. и

$= \frac{2-2\theta+2\theta}{\theta(1-\theta)}$. >0 на $\theta\in(0,\frac{1}{2})$

✓

$\Rightarrow$ модель регулярна.

$\tilde\theta_1, \tilde\theta_2$ – несмещ., $D\tilde\theta_1, D\tilde\theta_2$ огр. на $\forall$ комп. из Θ $\Rightarrow$ они явл. регулярн. оценками.

$\Rightarrow$ Выполн. неравенство Кр.-Р.

$$\cancel{D\tilde{g}(\vec{X}_n)} \tilde{\theta} \geq \frac{\theta(1-\theta)}{n \cdot 2}$$

$$D\tilde{\theta}_1 = \frac{45}{24n}\theta - \frac{1}{n}\theta^2$$

$$D\tilde{\theta}_2 = \frac{\theta(1-\theta)}{n}$$

/ ничего нельзя сказать об эффект.